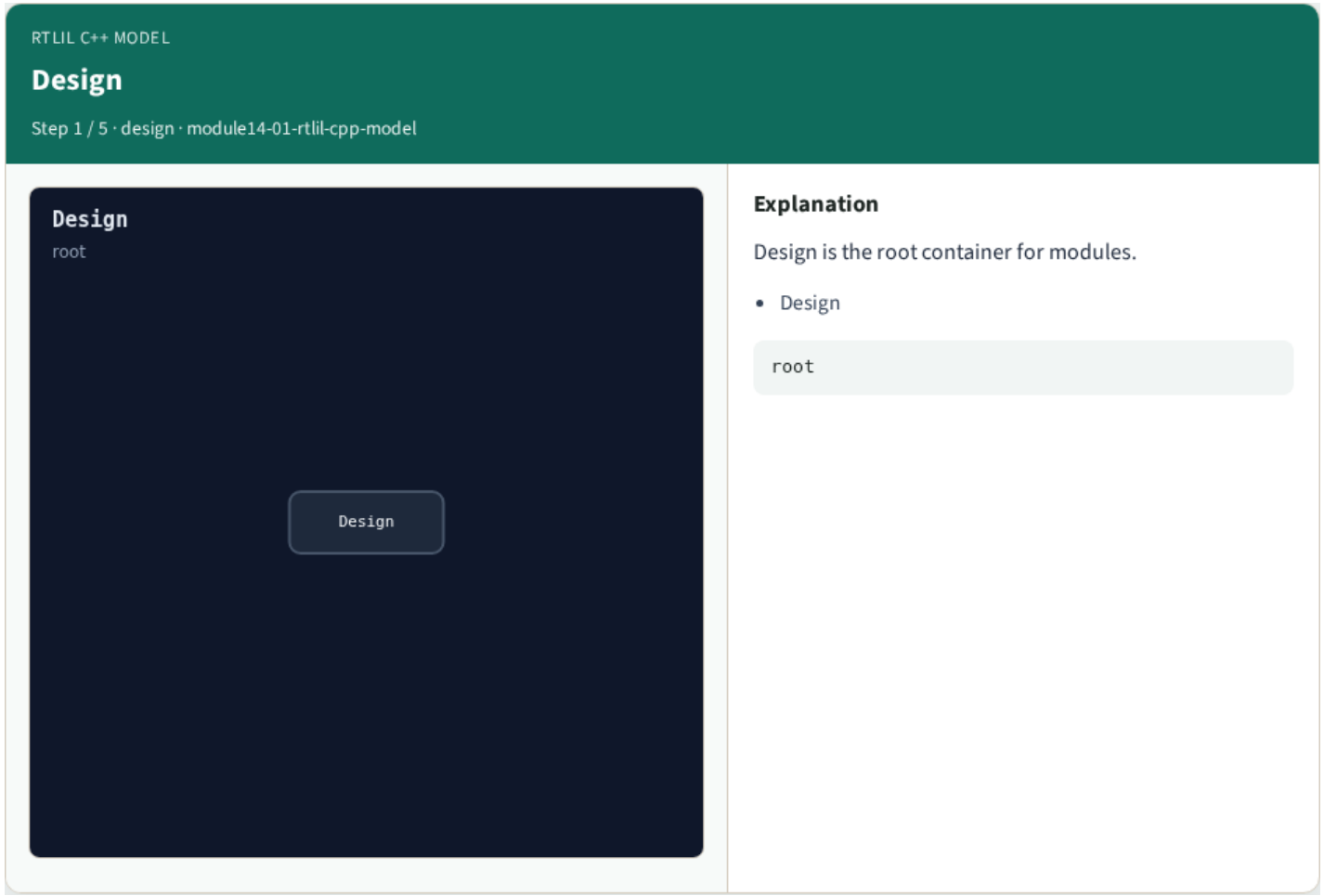
Why RTLIL C++

Browser IR and JSON are maps

The idea

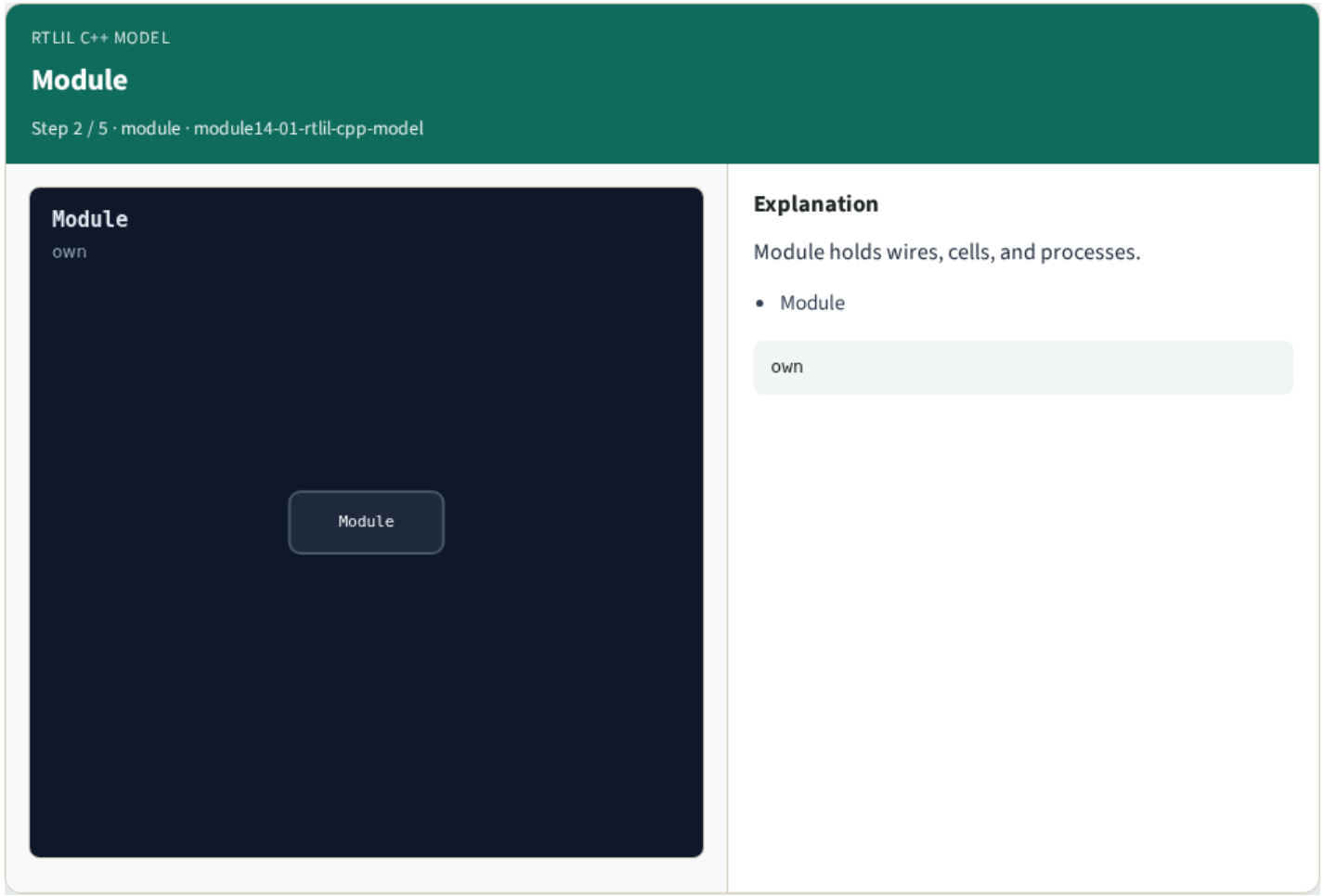
- GOLDEN: Design owns Modules; Module owns Wires, Cells, Processes; ports use SigSpec

Design



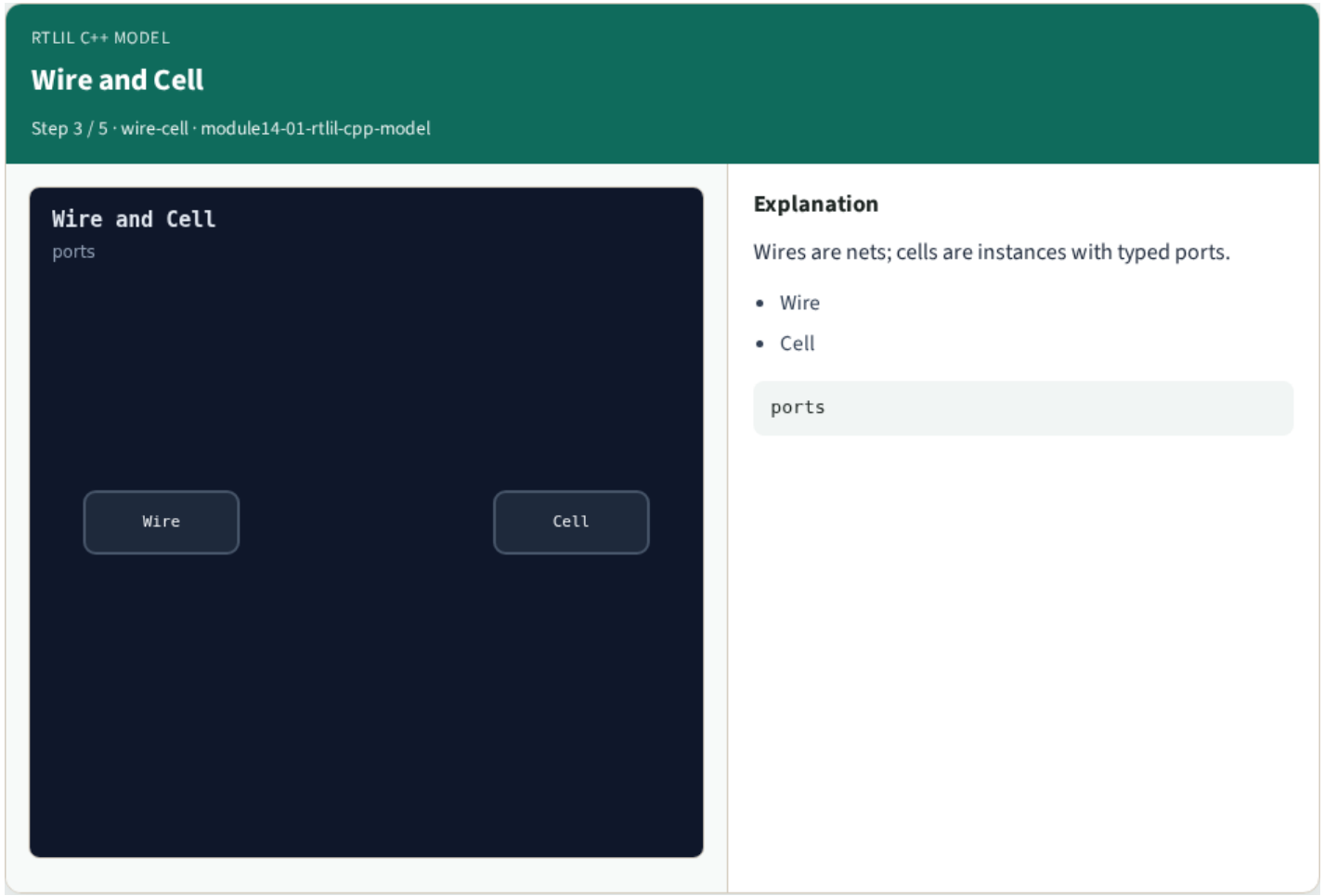
Design

Module



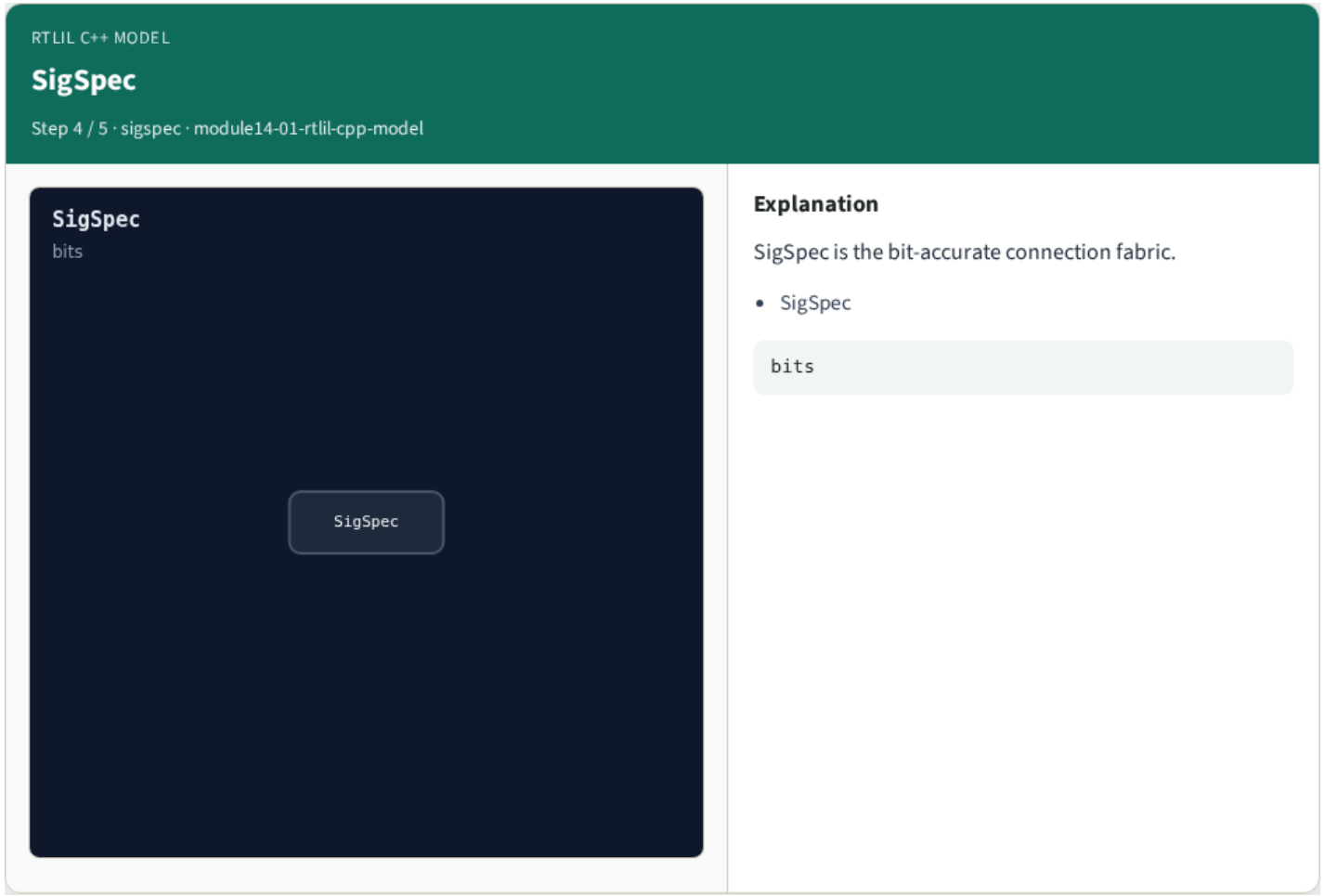
Module

Wire and Cell



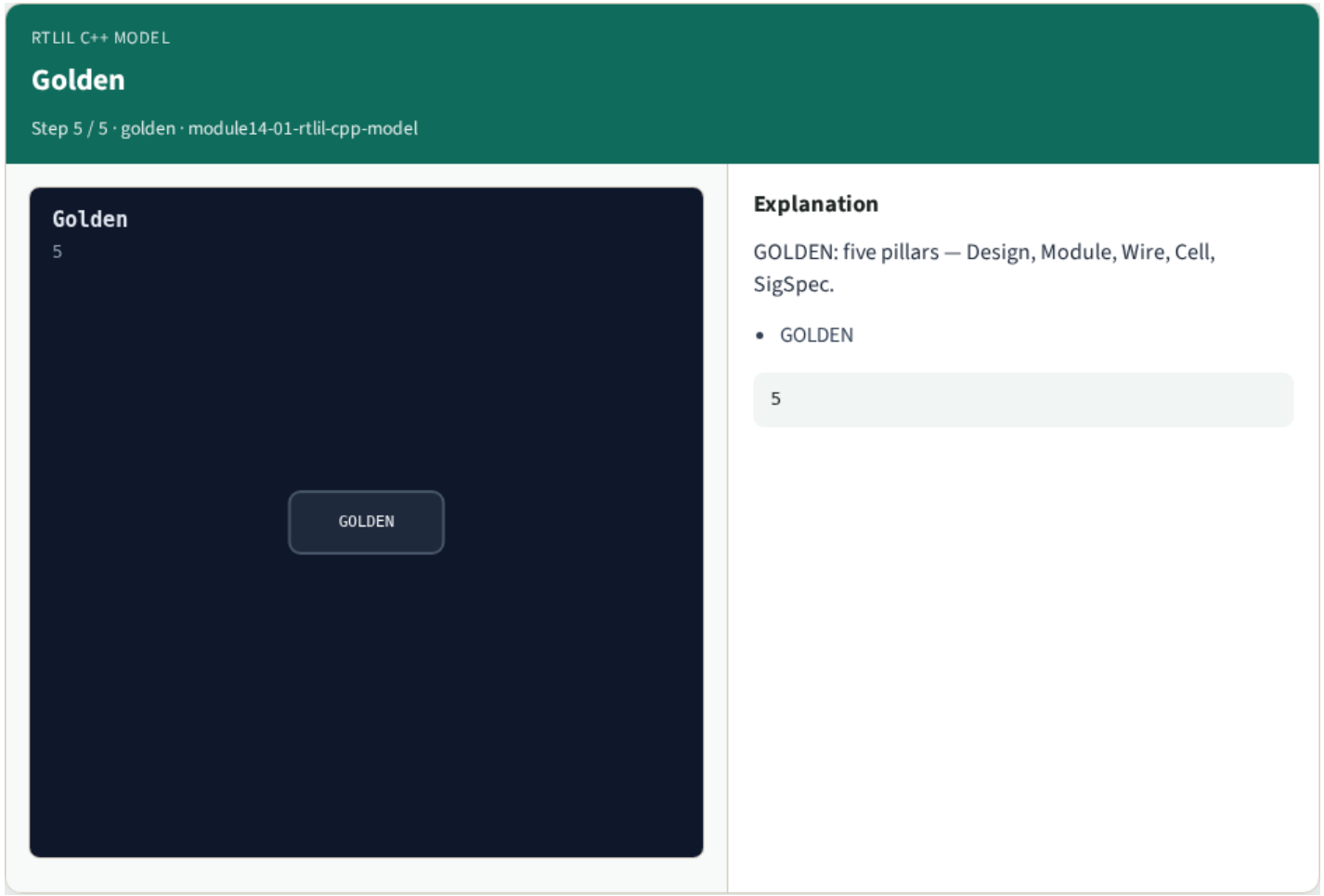
Wire and Cell

SigSpec



SigSpec

Golden



Golden

Browser lab track

- Open **rtlil-cpp-model**
- Click through the object graph and check the goldens

Implement track

- Call rtlil_cpp_model() in internals_solvers; assert pillar count and nesting

Pitfalls

- Treating JSON keys as the C++ API
- Forgetting SigSpec between cells and wires

Try these

- From the common folder, run the internals solvers self-test

Try these — try these

```
cd courses/learn_yosys/common
python3 internals_solvers.py
```

Pseudocode

- Here is the RTLIL nesting in plain steps

Pseudocode — try these

```
design.modules[name] → module
module.wires / module.cells / module.processes
cell.connections[port] → SigSpec
```

Next

- Complete the quiz for this part
- Next: how passes register and run on a Design